



**中國海洋大學海德學院**

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

# **Statistical Modelling and Inference**

**Week 4, Lecture 1**

**Outline:**

- i. Simple Linear Regression**
- ii. Least Squares Estimation**

# Simple Linear Regression

- Our objective is to study the relationship between two variables  $X$  and  $Y$ .
- One way to study the relationship between two variables is by using of **regression**.
- Regression analysis is the process of estimating a functional relationship between  $X$  and  $Y$ . A regression equation is often used to predict a value of  $Y$  for a given value of  $X$ .
- Another way to study relationship between two variables is **correlation**. It involves measuring the direction and the strength of the **linear** relationship.

# First-Order Linear Model or Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where

**y = dependent variable**

**x = independent variable**

**$\beta_0$  = y-intercept**

**$\beta_1$  = slope of the line**

**$\varepsilon$  = error variable**

# Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

This model is:

- i. **Simple:** only one  $X$
- ii. **Linear in the parameters:** No parameter appears as exponent or is multiplied or divided by another parameter.
- iii. **Linear in the predictor variable ( $X$ ):**  $X$  appears only in the first power.

# Examples of Regression Models

- Multiple Linear Regression:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

- Polynomial Linear Regression:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

- Linear Regression:

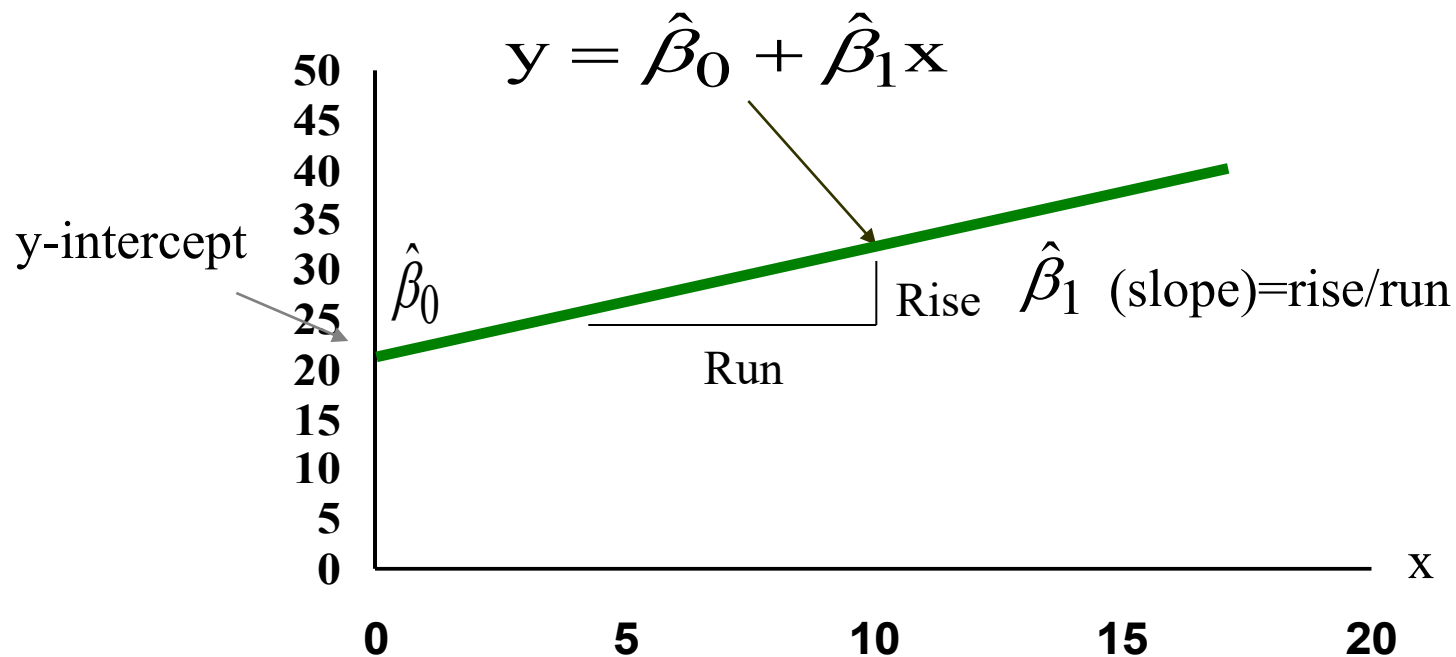
$$\log_{10}(Y_i) = \beta_0 + \beta_1 X_i + \beta_2 \exp(X_i) + \varepsilon_i$$

- Nonlinear Regression:

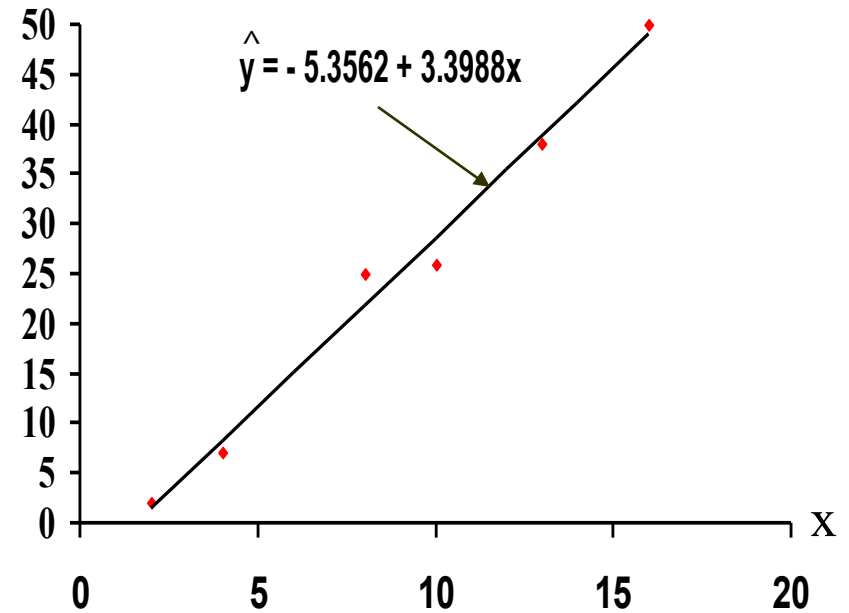
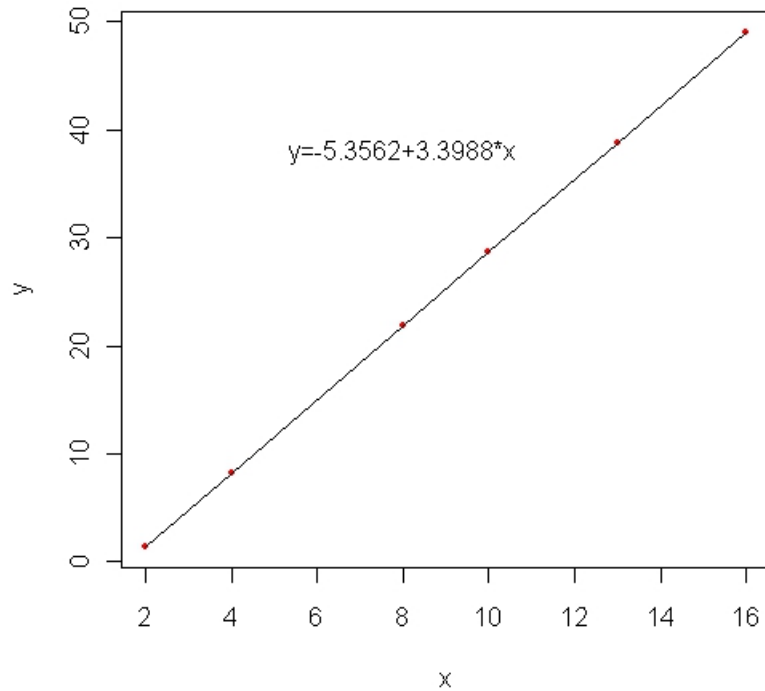
$$Y_i = \beta_0 / (1 + \beta_1 \exp(\beta_2 X_i)) + \varepsilon_i$$

note: nonlinear in parameters

# Components of a Simple Linear Regression Model



# Mathematical vs Statistical Relation



# Error

- **The scatterplot shows that the points are not on a line, and so, in addition to the relationship, we also describe the error:**

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i=1,2,\dots,n$$

- The  $y$ 's are the response (or dependent) variable. The  $x$ 's are the predictors or independent variables, and the epsilon's are the errors. We assume that the errors are normal, mutually independent, and have variance  $\sigma^2$ .



# Least Squares Estimation:

Minimize 
$$\sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The minimizing y-intercept and slope are given by  $\hat{\beta}_0$  and  $\hat{\beta}_1$ . We use the notation:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- The quantities  $R_i = y_i - \hat{y}_i$  are called the residuals. If we assume a normal error, they should look normal.

# Least Squares Estimation:

- The Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- The Least Squares Regression Line,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_x}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$SS_x = \sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}$$

$$SS_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n}$$

# What form does the error take?

- Each observation may be decomposed into two parts:

$$y = \hat{y} + (y - \hat{y})$$

- The first part is used to determine the fit, and the second to estimate the error.
- We estimate the standard deviation of the error by:

$$SSE = \sum (Y - \hat{Y})^2 = S_{yy} - \left( \frac{S_{xy}^2}{S_{xx}} \right)$$

# Estimation of $\sigma^2$ .

- We estimate  $\sigma^2$  by

$$s^2 = \frac{SSE}{n - 2} = MSE$$

where,

$$SSE = \sum (Y - \hat{Y})^2 = S_{yy} - \left( \frac{S_{xy}^2}{S_{xx}} \right)$$

## Example:

An educational economist wants to establish the relationship between an individual's income and education. He takes a random sample of 10 individuals and asks for their income (in \$1000s) and education (in years). The results are shown below. Find the least squares regression line.

|                  |    |    |    |    |    |    |    |    |    |    |
|------------------|----|----|----|----|----|----|----|----|----|----|
| <b>Income</b>    | 25 | 33 | 22 | 41 | 18 | 28 | 32 | 24 | 53 | 26 |
| <b>Education</b> | 11 | 12 | 11 | 15 | 8  | 10 | 11 | 12 | 17 | 11 |

# Dependent and Independent Variables

- The dependent variable is the one that we want to forecast or analyze ( $y$  income).
- The independent variable is hypothesized to affect the dependent variable ( $x$  education).
- In this example, we wish to analyze income, and we choose the variable individual's education that most effects income. Hence,  $y$  is income and  $x$  is individual's education.

$$\sum x_i = 118$$

$$\sum x_i^2 = 1450$$

$$\sum y_i = 302$$

$$\sum y_i^2 = 10072$$

$$\sum x_i y_i = 3779$$

$$SS_x = \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 1450 - \frac{(118)^2}{10} = 57.6$$

$$SS_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 3779 - \frac{(118)(302)}{10} = 215.4$$

# Least Squares Estimation:

Now slope

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_x} = \frac{215.4}{57.6} = 3.74$$

and intercept

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{302}{10} - 3.74 \frac{118}{10} = -13.93$$

Therefore, The least squares regression line is;

$$\hat{y} = -13.93 + 3.74x$$



# The Least Squares Regression Line

- The least squares regression line is

$$\hat{y} = -13.93 + 3.74x$$

- Interpretation of coefficients:

The sample slope  $\hat{\beta}_1 = 3.74$  tells us that on average for each additional year of education, an individual's income rises by \$3.74 thousand.

- The y-intercept is  $\hat{\beta}_0 = -13.93$ . This value is the expected (or average) income for an individual who has 0 education level (which is meaningless here)

# Error Variable:

- $\varepsilon$  is normally distributed.
- $E(\varepsilon) = 0$
- The variance of  $\varepsilon$  is  $\sigma^2$ .
- The errors are independent of each other.
- The estimator of  $\sigma^2$  is

$$s_{\varepsilon}^2 = \frac{SSE}{n-2} = MSE$$

where;

$$SSE = \sum (Y - \hat{Y})^2 = S_{yy} - \left( \frac{S_{xy}^2}{S_{xx}} \right) \text{ and } SS_y = \sum y_i^2 - \frac{(\sum y_i)^2}{n}$$

## Example (continue)

- For the previous example

$$SS_y = \sum y_i^2 - \frac{(\sum y_i)^2}{n} = 10072 - \frac{(302)^2}{10} = 951.6$$

Hence, SSE is

$$SSE = S_{yy} - \left( \frac{S_{xy}^2}{S_{xx}} \right) = 951.6 - \frac{(215.4)^2}{57.6} = 146.09$$

$$s_\varepsilon^2 = \frac{SSE}{n-2} = \frac{146.09}{10-2} = 18.26 \text{ and } s_\varepsilon = \sqrt{s_\varepsilon^2} = 4.27$$

# Interpretation of $s_{\varepsilon}^2$

- The value of “s” can be compared with the mean value of  $y$  to provide a rough guide as to whether  $s$  is small or large.
- Since  $\bar{y} = 30.2$  and  $s_{\varepsilon} = 4.27$ , we would conclude that “s” is relatively small, indicating that the regression line fits the data quite well or best fitted regression line.

## **Example:**

- Car dealers across North America use the red book to determine a cars selling price on the basis of important features. One of these is the car's current odometer reading.
- To examine this issue 100 three-year old cars in mint condition were randomly selected. Their selling price and odometer reading were observed.

# Portion of the data file

| <b>Odometer</b> | <b>Price</b> |
|-----------------|--------------|
| 37388           | 5318         |
| 44758           | 5061         |
| 45833           | 5008         |
| 30862           | 5795         |
| .....           | ...          |
| 34212           | 5283         |
| 33190           | 5259         |
| 39196           | 5356         |
| 36392           | 5133         |

# Example (R Output)

## Regression Analysis

The regression equation is

**Price = 6533 - 0.0312 Odometer**

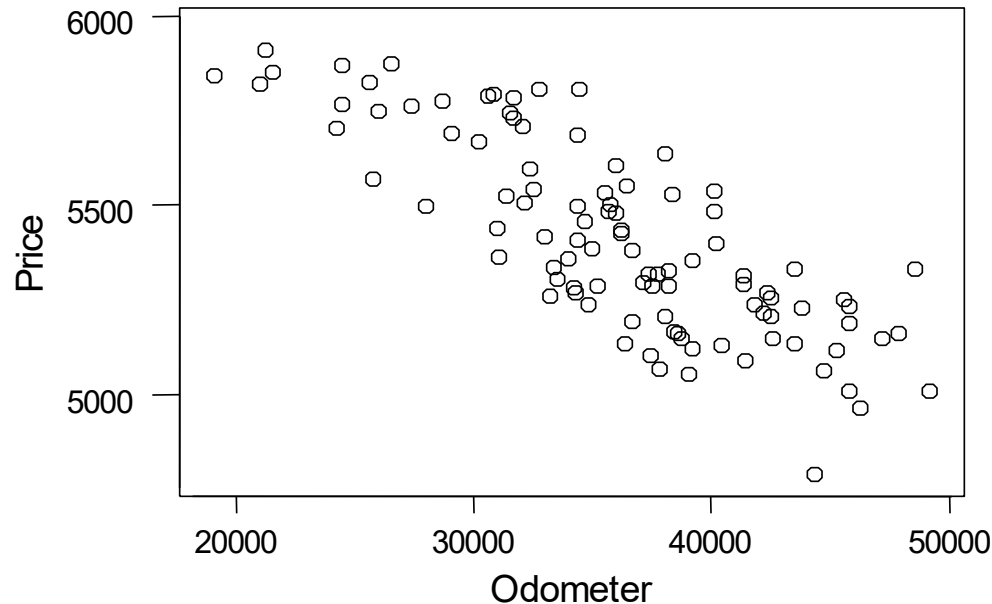
| Predictor | Coef      | StDev    | T      | P                   |
|-----------|-----------|----------|--------|---------------------|
| Constant  | 6533.38   | 84.51    | 77.31  | 0.000 (SIGNIFICANT) |
| Odometer  | -0.031158 | 0.002309 | -13.49 | 0.000 (SIGNIFICANT) |

S = 151.6                  R-Sq = 65.0%                  R-Sq(adj) = 64.7%

# Example

- The least squares regression line is

$$\hat{y} = 6533.38 - 0.031158x$$

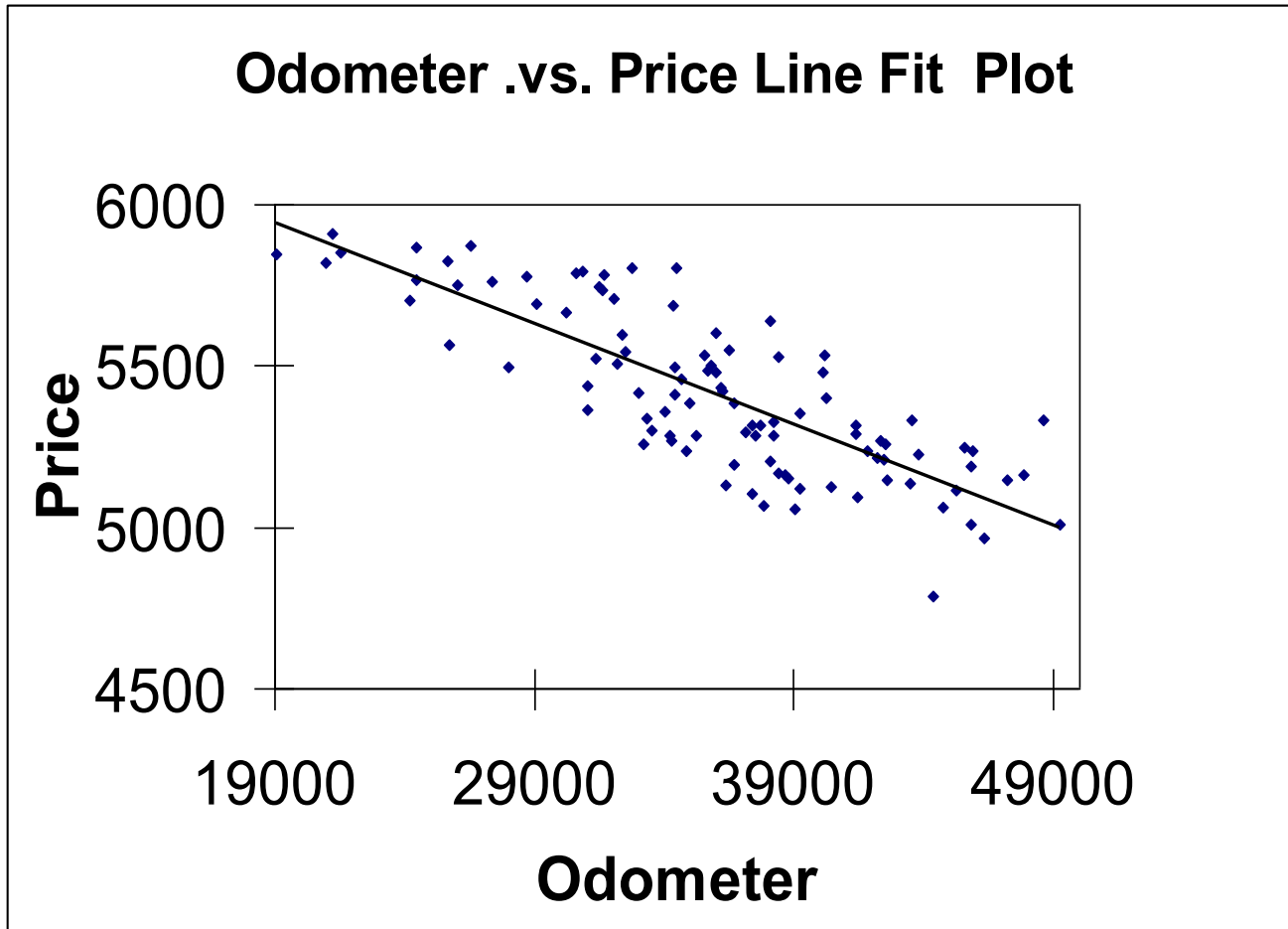




# Interpretation of the coefficients

- $\hat{\beta}_1 = -0.031158$  means that for each additional mile on the odometer, the price decreases by an average of 3.1158 cents.
- $\hat{\beta}_0 = 6533.38$  means that when  $x = 0$  (new car), the selling price is \$6533.38 but  $x = 0$  is not in the range of  $x$ . So, we cannot interpret the value of  $y$  when  $x=0$  for this problem.
- $R^2=65.0\%$  means that 65% of the variation of  $y$  can be explained by  $x$ . The higher the value of  $R^2$ , the better the model fits the data.

# Scatter Plot



# Testing the slope

- Are X and Y linearly related?

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

- Test Statistic:

$$t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} \quad \text{Where; } s_{\hat{\beta}_1} = \frac{s_{\varepsilon}}{\sqrt{SS_x}}$$

# Testing the slope (continue)

- The Rejection Region:

Reject  $H_0$  if  $t < -t_{\alpha/2, n-2}$  or  $t > t_{\alpha/2, n-2}$ .

- If we are testing that high  $x$  values lead to high  $y$  values,  $H_A: \beta_1 > 0$ . Then, the rejection region is  $t > t_{\alpha, n-2}$ .
- If we are testing that high  $x$  values lead to low  $y$  values or low  $x$  values lead to high  $y$  values,  $H_A: \beta_1 < 0$ . Then, the rejection region is  $t < -t_{\alpha, n-2}$ .

# Assessing the model

Example R output:

| Predictor | Coef      | StDev    | t      | P     |
|-----------|-----------|----------|--------|-------|
| Constant  | 6533.38   | 84.51    | 77.31  | 0.000 |
| Odometer  | -0.031158 | 0.002309 | -13.49 | 0.000 |

# Using the Regression Equation

- Suppose we would like to predict the selling price for a car with 40,000 miles on the odometer

$$\begin{aligned}\hat{y} &= 6,533 - 0.0312x \\ &= 6,533 - 0.0312(40,000) \\ &= \$5,285\end{aligned}$$

# Prediction and Confidence Intervals

- **Prediction Interval of  $y$  for  $x=x_g$ :** The confidence interval for predicting the particular value of  $y$  for a given  $x$

$$\hat{y} \pm t_{\alpha/2, n-2} S_e \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{SS_x}}$$

- **Confidence Interval of  $E(y|x=x_g)$ :** The confidence interval for estimating the expected value of  $y$  for a given  $x$

$$\hat{y} \pm t_{\alpha/2, n-2} S_e \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{SS_x}}$$

# Prediction Interval

- From previous calculations we have the following estimates:

$$\hat{y} = 5285, s_{\varepsilon} = 151.6, SS_x = 4309340160, \bar{x} = 36,009$$

- Thus a 95% **prediction interval** for  **$x=40,000$**  is:

$$5,285 \pm 1.984(151.6) \sqrt{1 + \frac{1}{100} + \frac{(40,000 - 36,009)^2}{4,309,340,160}}$$

$$5,285 \pm 303$$

The prediction is that the selling price of the car will fall between **\$4982** and **\$5588**.



# Confidence Interval

- A 95% confidence interval of

$E(y | x=40,000)$  is:

$$5,285 \pm 1.984(151.6) \sqrt{\frac{1}{100} + \frac{(40,000 - 36,009)^2}{4,309,340,160}}$$

$$5,285 \pm 35$$

$$(5250, 5320)$$

The mean selling price of the car will fall between \$5250 and \$5320.

# Exercise

- It is doubtful that any sports collects more statistics than baseball. The fans are always interested in determining which factors lead to successful teams. The table below lists the team batting average and the team winning percentage for the 14 league teams at the end of a recent season.

| Team-B-A | Winning% |
|----------|----------|
| 0.254    | 0.414    |
| 0.269    | 0.519    |
| 0.255    | 0.500    |
| 0.262    | 0.537    |
| 0.254    | 0.352    |
| 0.247    | 0.519    |
| 0.264    | 0.506    |
| 0.271    | 0.512    |
| 0.280    | 0.586    |
| 0.256    | 0.438    |
| 0.248    | 0.519    |
| 0.255    | 0.512    |
| 0.270    | 0.525    |
| 0.257    | 0.562    |

$y = \text{winning \%}$  and  $x = \text{team batting average}$

## a) LS Regression Line

$$\sum x_i = 3.642, \sum x_i^2 = 0.949$$

$$\sum y_i = 7.001, \sum y_i^2 = 3.549$$

$$\sum x_i y_i = 1.824562$$

$$SS_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 1.824562 - \frac{(3.642)(7.001)}{14} = 0.0033$$

$$SS_x = \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 0.948622 - \frac{(3.642)^2}{14} = 0.00118$$

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_x} = \frac{0.003302}{0.001182} = 0.7941$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.5 - (0.7941)0.26 = 0.2935$$

- The least squares regression line is

$$\hat{y} = 0.2935 + 0.7941x$$

- The meaning of  $\hat{\beta}_1 = 0.7941$  is, when the batting average of the team increase, the winning percentage increases. Or positive relationship between batting average and winning percentage.

## b) Standard Error of Estimate

$$\begin{aligned}SSE &= S_{yy} - \left( \frac{S_{xy}^2}{S_{xx}} \right) = \left( \sum y_i^2 - \frac{(\sum y_i)^2}{n} \right) - \left( \frac{S_{xy}^2}{S_{xx}} \right) \\&= \left( 3.548785 - \frac{7.001^2}{14} \right) - \frac{0.003302^2}{0.00182} = 0.03856\end{aligned}$$

$$\text{So, } s_{\varepsilon}^2 = \frac{SSE}{n-2} = \frac{0.03856}{14-2} = 0.00321 \text{ and } s_{\varepsilon} = \sqrt{s_{\varepsilon}^2} = 0.0567$$

- Since  $s_{\varepsilon}=0.0567$  is small, we would conclude that “s” is relatively small, indicating that the regression line fits the data quite well.

c) Do the data provide sufficient evidence at the 5% significance level to conclude that higher team batting average lead to higher winning percentage?

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 > 0$$

$$\text{use } s_{\hat{\beta}_1} = \frac{s_{\varepsilon}}{\sqrt{SS_x}}$$

$$\text{Test statistic: } t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} = 1.69 \quad (\text{p-value}=.058)$$

**Conclusion:** Do not reject  $H_0$  at  $\alpha = 0.05$  and conclude that the higher team batting average does not lead to higher winning percentage.

d) Predict with 90% confidence interval for the winning percentage of a team whose batting average is 0.275.

$$\hat{y} = 0.2935 + 0.7941(0.275) = 0.5119$$

$$\hat{y} \pm t_{\alpha/2, n-2} s_{\varepsilon} \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{SS_x}} =$$

$$0.5119 \pm (1.782)(0.0567) \sqrt{1 + \frac{1}{14} + \frac{(0.275 - 0.2601)^2}{0.001182}}$$

$$0.5119 \pm 0.1134$$

90% PI for y: (0.3985, 0.6253)

- The prediction is that the winning percentage of the team will fall between 39.85% and 62.53%.



**Thank  
You**

**谢谢**

